

IFRS 9 - impairment - compliance audit

IFRS 9 Financial Instruments - impairment (5.5 and B5.5): staging, significant increase in credit risk, 12-month and lifetime expected credit losses, forward-looking information, write-off.

Malawi Agricultural and Industrial Investment Company (MAIIC) - IFRS 9 EIR and ECL system - build of 8 October 2026 against specification v4. Prepared 8 October 2026 from the build of specification v4: the staged loan books of July 2024 to August 2026 on the clean install, the Governance Centre's staging and forward-looking settings, the forward-looking chain (fli:correlate), the scenario set of 15.8, and the PHPUnit suite. The PD engine (transition matrices) has not run on the clean install, so the ECL figures there are nil; the rows below audit the mechanics and the settings. Reviewer: Dr Thomson Kumwenda (CFO), then Deloitte. The Excel workbook of the same name is the working copy (drop-down statuses, hyperlinked contents, reviewer sign-off column, a live Baselines sheet); this document is its printable twin.

Source: IFRS 9 Financial Instruments as issued by the IASB, section 5.5 and B5.5.1 to B5.5.55. The Reserve Bank of Malawi Financial Services (Credit Risk Management for Development Finance Institutions) Directive 2018 (docs/regulatory). Specification: docs/MAIIC_EIR_Engine_Specification_v4_2026-10-07.md sections 3.6, 13, 14, 15, 16.

Status summary

Status	Meaning	Sections
Done	Implemented in the system and visible in an approved output; the test in column 11 fails if the behaviour changes.	9
Partially done	The mechanics are in place, but a parameter, an input or a piece of evidence still differs from the requirement.	2
Outstanding	Required for MAIIC and not built.	0
Not applicable, documented	Does not apply to MAIIC by a recorded decision or fact (reason and reference given); revisit if the facts change.	0
Evidence needed from MAIIC	The system is ready; MAIIC must supply a document, minute or dataset before the row can be closed.	3
Total		14

Contents

Ref	Section	Status	Governance setting	Test
5.5 RECOGNITION OF EXPECTED CREDIT LOSSES				
5.5.1	A loss allowance for expected credit losses	Done		Tests\Feature\Ecl\Time PhasedEclServiceTest; Tests\Feature\Ecl\EclDiscountRateServiceTest
5.5.3	Lifetime ECL on a significant increase in credit risk	Done	dpd_basis	Tests\Feature\Eir\StagingServiceTest
5.5.5	12-month ECL where credit risk has not increased significantly	Done		Tests\Feature\Ecl\Time PhasedEclServiceTest
5.5.9 to 5.5.11	Determining a significant increase in credit risk; the 30-days-past-due rebuttable presumption	Done	staging_rebuttal; dpd_basis	Tests\Feature\Eir\StagingThresholdSeederTest; Tests\Feature\Eir\StagingServiceTest

Ref	Section	Status	Governance setting	Test
B5.5.37	Definition of default; the 90-days-past-due rebuttable presumption	Done	dpd_basis; stage3_missed_installments; staging_rebuttal	Tests\Feature\Eir\StagingServiceTest::test_the_directive_thresholds_by_tenor_class_and_the_bucket_fallback
5.5 MEASUREMENT				
5.5.17	Measurement of expected credit losses	Partially done	scenario_weighting_method; fli_adjustment_route	Tests\Feature\Scenario\ScenarioSetServiceTest; Tests\Feature\Ecl\EclDiscountRateServiceTest
B5.5.41 to B5.5.43	Probability-weighted outcomes	Done	scenario_minimum_count; scenario_weight_bounds; scenario_calibration_note	Tests\Feature\Scenario\ScenarioSetServiceTest
B5.5.49 to B5.5.54	Reasonable and supportable information; forward-looking information	Done	fli_expected_sign_test; fli_r2_cutoff; fli_min_observations; fli_alpha; macro_source_precedence	Tests\Feature\FLI\FliBridgeServiceTest
B5.5.52 (transmission)	How forward-looking information reaches the PD	Done	fli_transmission_method; fli_asset_correlation	Tests\Feature\FLI\TransmissionMethodCatalogueTest
B5.5.28 to B5.5.30	Collective assessment and segmentation	Partially done	fli_transmission_method	Tests\Feature\FLI\FliBridgeServiceTest
B5.5.33 to B5.5.35	Loss given default and collateral	Evidence needed from MAIC		
5.5.17(b), B5.5.44	Time value of money	Done		Tests\Feature\Ecl\EclDiscountRateServiceTest
5.5 WRITE-OFF AND PRESENTATION				
5.4.4, B5.4.9	Write-off	Evidence needed from MAIC		
THE MEGA FARM PROGRAMME				
16	The Mega Farm loans in the allowance	Evidence needed from MAIC	mega_farms_scope; megafarm_pd_method; megafarm_scalar_ceiling	

Baselines - the acceptance ties read from the system

Generated 2026-10-08 20:53:34 from maiic_ifrs9_bootstrap.

Tie	Expected	System figure	Result
Interest posted Jan 2025 to Jul 2026 (ledger, types 303 and 120)	5,293,988,207.06	5,293,988,207.06	PASS
Derived carrying amount against the stored report (over 2 cents)	0 differences	0 of 2361 account-months	PASS
Year-end interest batch contra on 4215	32,956,675.55	32,956,675.55	PASS
Year-end interest batch contra on 4216	-21,733,262.70	-21,733,262.70	PASS
Loan book 1050101 at 31 Dec 2025 against the trial balance	1,368,737,810.68	1,368,737,810.68	PASS

Tie	Expected	System figure	Result
Loan book 1050102 at 31 Dec 2025 against the trial balance	6,434,045,052.56	6,434,045,052.56	PASS
Loan book 1050201 at 31 Dec 2025 against the trial balance (accepted exception, spec 3.5: -400,000.00)	1,402,393,139.46	1,401,993,139.46	PASS
Loan book 1050202 at 31 Dec 2025 against the trial balance (accepted exception, spec 3.5: 1,000,000.00)	3,808,288,656.15	3,809,288,656.15	PASS
Loan book 1050401 at 31 Dec 2025 against the trial balance	1,827,148,120.69	1,827,148,120.69	PASS

Findings - items that need a decision or a fix

F1. The PD and LGD of the clean install are the bootstrap's, not MAIIC's (5.5.17) - Open

- **What was found:** PdEngineService and LgdEngineService run the transition matrix and the cohort workout under the bootstrap label; no MAIIC reviewer has key-locked a matrix or an LGD calculation in the system.
- **Impact:** The ECL of 8.33bn at August 2026 is a system figure awaiting review.
- **Recommended action:** A reviewer key-locks the matrix and the LGD on their screens; the ECL is then re-run under the approved scenario set.
- **Owner:** Dr Thom / the reviewer

F2. No fit is approved by a MAIIC reviewer (B5.5.49) - Open

- **What was found:** The chain applied 39 fits by the guardrail; approval of a fit for use in the FLI route is a maker-checker decision not yet taken.
- **Impact:** The forward-looking adjustment is the manual overlay at zero until then.
- **Recommended action:** A reviewer approves a fit on the FLI Adjustments screen; a second person reviews.
- **Owner:** Dr Thom / the reviewer

F3. The Mega Farm provision basis is undocumented in the system (16) - Open

- **What was found:** The 2025 statements carry K39.76 bn against K48.7 bn Stage 3; the November 2025 book in the system is mostly Stage 1 by days past due.
- **Impact:** The ECL golden number at 31 December 2025 cannot be set.
- **Recommended action:** The CFO confirms D30 and the provisioning method; the monthly Mega Farm books are loaded by the feed (MF_01 to MF_08).
- **Owner:** Dr Thom / Barry

Section-by-section audit

5.5 RECOGNITION OF EXPECTED CREDIT LOSSES

5.5.1. A loss allowance for expected credit losses - Done

- **What it requires:** Recognise a loss allowance for expected credit losses on a financial asset measured at amortised cost.
- **What the engine does:** TimePhasedEclService computes the allowance per loan as EAD x PD x LGD, discounted at the EIR, for the period; ifrs9:recalculate-ecl runs it per portfolio; the bootstrap runs it for the last period.
- **Compliance comment:** Compliant in mechanics; the PD engine must run for a figure.
- **Where to see it:** Financial Modelling, ECL Calculation: /expected-credit-loss

- **Test that proves it:** Tests\Feature\Ecl\TimePhasedEclServiceTest;
Tests\Feature\Ecl\EclDiscountRateServiceTest
- **Reviewer sign-off:** _____ Date: _____

5.5.3. Lifetime ECL on a significant increase in credit risk - *Done*

- **What it requires:** At each reporting date, measure the loss allowance at lifetime ECL if the credit risk has increased significantly since initial recognition.
- **What the engine does:** StagingService stages every period: Stage 2 from 31 days past due in every class (the directive), the SICR flag from the qualitative groups, and the ECL takes the lifetime window for Stage 2.
- **General comment:** Spec v4 section 3.6, decision D31.
- **Compliance comment:** Compliant.
- **Where to see it:** Governance Centre, Staging & SICR Rules: /stageing-rules; Data Foundation, Loan Book
- **Governance setting:** dpd_basis
- **Test that proves it:** Tests\Feature\Eir\StagingServiceTest
- **Reviewer sign-off:** _____ Date: _____

5.5.5. 12-month ECL where credit risk has not increased significantly - *Done*

- **What it requires:** Measure the loss allowance at an amount equal to 12-month ECL where the credit risk has not increased significantly since initial recognition.
- **What the engine does:** Stage 1 loans take the 12-month PD window in the ECL service; the staging writes the stage every period.
- **Compliance comment:** Compliant.
- **Where to see it:** Financial Modelling, ECL Calculation: /expected-credit-loss
- **Test that proves it:** Tests\Feature\Ecl\TimePhasedEclServiceTest
- **Reviewer sign-off:** _____ Date: _____

5.5.9 to 5.5.11. Determining a significant increase in credit risk; the 30-days-past-due rebuttable presumption - *Done*

- **What it requires:** Assess the change in the risk of default since initial recognition using reasonable and supportable information; there is a rebuttable presumption that credit risk has increased significantly when payments are more than 30 days past due.
- **What the engine does:** Stage 2 from 31 days past due in every facility class; days counted from the oldest overdue instalment as the directive counts, with the bucket ageing beside it; a rebuttal of the presumption is a governed setting (staging_rebuttal) whose seeded value cites the directive; the LONG_TERM rule proposing 91 days is seeded future-dated and inactive pending CFO sign-off.
- **General comment:** The O13 rebuttal paper of 8 October.
- **Compliance comment:** Compliant; the presumption is applied, the rebuttal is governed.
- **Where to see it:** Governance Centre, Staging & SICR Rules: /stageing-rules; Data Foundation, Loan Book; Governance Centre: /eir-governance
- **Governance setting:** staging_rebuttal; dpd_basis
- **Test that proves it:** Tests\Feature\Eir\StagingThresholdSeederTest;
Tests\Feature\Eir\StagingServiceTest
- **Reviewer sign-off:** _____ Date: _____

B5.5.37. Definition of default; the 90-days-past-due rebuttable presumption - *Done*

- **What it requires:** Apply a definition of default consistent with the one used for internal credit risk management; there is a rebuttable presumption that default occurs no later than 90 days past due unless reasonable and supportable information supports a more lagging criterion.
- **What the engine does:** Stage 3 from 91 days for a facility of 12 months or less and for the Mega Farm class, and from 181 days for a medium- or long-term facility, as the RBM directive classifies them; the accounting policy's four-consecutive-missed-instalments trigger lifts a loan to Stage 3 beside the day count. The more

lagging criterion rests on the directive, cited on each threshold row.

- **General comment:** Spec v4 section 3.6 compares the policy (90 days), the November 2025 model (181) and the directive (181/91).
- **Compliance comment:** Compliant; the rebuttal is documented on the threshold rows.
- **Where to see it:** Governance Centre, Staging & SICR Rules: /stageing-rules; Data Foundation, Loan Book
- **Governance setting:** dpd_basis; stage3_missed_instalments; staging_rebuttal
- **Test that proves it:** Tests\Feature\Eir\StagingServiceTest::test_the_directive_thresholds_by_ten_or_class_and_the_bucket_fallback
- **Reviewer sign-off:** _____ Date: _____

5.5 MEASUREMENT

5.5.17. Measurement of expected credit losses - *Partially done*

- **What it requires:** ECL is an unbiased, probability-weighted amount determined by evaluating a range of possible outcomes, the time value of money, and reasonable and supportable information about past events, current conditions and forecasts of future economic conditions.
- **What the engine does:** Probability weighting across scenarios is built as the governed weighting method (the ECL under each scenario, weighted; sensitivity stored with the set); discounting is at the EIR; the forward-looking chain produces fits from the macro series. The PD engine has not run on the clean install and no fit is approved by a MAIIC reviewer, so the weighted ECL is not yet produced for a period.
- **General comment:** Spec v4 sections 14, 15.
- **Compliance comment:** Partially done: the mechanics are in place; the PD run and an approved fit are awaited.
- **Where to see it:** Financial Modelling, ECL Calculation: /expected-credit-loss; Governance Centre, Scenario Sets: /scenario-sets
- **Governance setting:** scenario_weighting_method; fli_adjustment_route
- **Test that proves it:** Tests\Feature\Scenario\ScenarioSetServiceTest;
Tests\Feature\Ecl\EclDiscountRateServiceTest
- **Reviewer sign-off:** _____ Date: _____

B5.5.41 to B5.5.43. Probability-weighted outcomes - *Done*

- **What it requires:** The estimate reflects the possibility that a credit loss occurs and the possibility that none occurs; a range of scenarios, at least two, weighted by their probabilities.
- **What the engine does:** The governed scenario set: three or more scenarios, weights summing to 100, the base at least 40 percent and no single weight above 60, every scenario but the base a shock on the base path; the first set (Base 50, Upside 15, Downside 25, Severe 10) is seeded as a proposal for the CFO, each anchored to a year Malawi has lived through.
- **General comment:** Spec v4 section 15.
- **Compliance comment:** Compliant.
- **Where to see it:** Governance Centre, Scenario Sets: /scenario-sets
- **Governance setting:** scenario_minimum_count; scenario_weight_bounds; scenario_calibration_note
- **Test that proves it:** Tests\Feature\Scenario\ScenarioSetServiceTest
- **Reviewer sign-off:** _____ Date: _____

B5.5.49 to B5.5.54. Reasonable and supportable information; forward-looking information - *Done*

- **What it requires:** Use information that is reasonably available without undue cost or effort, including forecasts of future economic conditions; where a statistical relationship is used it must be supportable.
- **What the engine does:** Eleven Malawi macro series from the World Bank with their provenance (source, address, fetched-at, who) and a committed snapshot; the forward-looking chain sweeps every driver against every credit-loss proxy over the lag grid under the guardrail (expected sign, R2 cutoff, p-value, minimum observations) and declines a fit that fails with the reason; the structural-events register (the 2023 devaluation, COVID, the policy-rate cycle, the E-Banker take-on) is read by the diagnostics.

- **General comment:** On the clean install: 209 fits, 39 applied by the guardrail, 170 declined.
- **Compliance comment:** Compliant; an applied fit is a proposal under maker-checker.
- **Where to see it:** Financial Modelling, Forward-Looking Model: /regression; fli:correlate; fli:method-cards; Data Foundation, Macro Statistics
- **Governance setting:** fli_expected_sign_test; fli_r2_cutoff; fli_min_observations; fli_alpha; macro_source_precedence
- **Test that proves it:** Tests\Feature\FLI\FliBridgeServiceTest
- **Reviewer sign-off:** _____ Date: _____

B5.5.52 (transmission). How forward-looking information reaches the PD - Done

- **What it requires:** The method of adjusting the PD for the forecast must be documented and supportable.
- **What the engine does:** Five governed transmission methods, each with a card: what it does, the formula, what it implies, its preconditions checked live against the data with the deciding figure, and a worked example on one loan; a method whose preconditions are not met cannot be selected; the scalar is seeded because the history cannot yet support the richer methods.
- **General comment:** Spec v4 section 14.7.
- **Compliance comment:** Compliant.
- **Where to see it:** Financial Modelling, Forward-Looking Model: /regression; fli:correlate; fli:method-cards (fli:method-cards; /fli/transmission-methods)
- **Governance setting:** fli_transmission_method; fli_asset_correlation
- **Test that proves it:** Tests\Feature\FLI\TransmissionMethodCatalogueTest
- **Reviewer sign-off:** _____ Date: _____

B5.5.28 to B5.5.30. Collective assessment and segmentation - Partially done

- **What it requires:** Where information is not available at the instrument level, assess on a collective basis grouped by shared credit-risk characteristics.
- **What the engine does:** Credit-loss proxies are produced for the book and per product group (NPL ratio, Stage 3 share, 12-month default rate), and the segment-specific scalar is a governed method; the PD transition matrices by grade run from their screen and are not yet in the bootstrap chain.
- **Compliance comment:** Partially done.
- **Where to see it:** Financial Modelling, Forward-Looking Model: /regression; fli:correlate; fli:method-cards; Financial Modelling, PD Model
- **Governance setting:** fli_transmission_method
- **Test that proves it:** Tests\Feature\FLI\FliBridgeServiceTest
- **Reviewer sign-off:** _____ Date: _____

B5.5.33 to B5.5.35. Loss given default and collateral - Evidence needed from MAIIC

- **What it requires:** The cash shortfalls reflect the cash flows expected from collateral and other credit enhancements that are part of the contractual terms.
- **What the engine does:** LGD runs from the LGD screens (monthly and cumulative); the security register (P2_11, 185 rows) is landed and 31 active accounts carry no security on it (the request to Credit of 7 October).
- **Compliance comment:** Evidence needed: the collateral for the 31 accounts; the valuations.
- **Where to see it:** Financial Modelling, LGD Model; Data Foundation, Collateral Register
- **Reviewer sign-off:** _____ Date: _____

5.5.17(b), B5.5.44. Time value of money - Done

- **What it requires:** Discount the expected credit losses at the EIR determined at initial recognition.
- **What the engine does:** EclDiscountRateService takes the locked EIR of the loan (the contractual rate where none is locked, and says so on the row); the bootstrap solves and locks the EIR before the ECL for this reason.
- **Compliance comment:** Compliant.

- **Where to see it:** Financial Modelling, ECL Calculation: /expected-credit-loss
- **Test that proves it:** Tests\Feature\Ecl\EclDiscountRateServiceTest
- **Reviewer sign-off:** _____ **Date:** _____

5.5 WRITE-OFF AND PRESENTATION

5.4.4, B5.4.9. Write-off - *Evidence needed from MAIIC*

- **What it requires:** Write off the gross carrying amount when there is no reasonable expectation of recovery.
- **What the engine does:** No write-off has been posted in E-Banker since the take-on (DD_16i returned no rows); the policy is MAIIC's to state.
- **Compliance comment:** Evidence needed: the write-off policy.
- **Where to see it:** Financial Modelling, ECL Calculation: /expected-credit-loss
- **Reviewer sign-off:** _____ **Date:** _____

THE MEGA FARM PROGRAMME

16. The Mega Farm loans in the allowance - *Evidence needed from MAIIC*

- **What it requires:** Loans extended under the Government's Mega Farm programme (losses to the fund; MAIIC's 5 percent share) must be in the ECL on a documented basis.
- **What the engine does:** The system holds a November 2025 Mega Farm loan book of 3,490 loans across six GLs (K54.85 bn), found by the build, staged under the MEGA_FARM class (91 days); the programme's treatment (decision D30: out of the EIR engine, in the ECL module, 5 percent share on net) and the PD method (seeded: the agricultural-sector PD scaled to the programme) are governed settings awaiting the CFO's confirmation.
- **General comment:** Spec v4 section 16; the O22 paper of 8 October.
- **Compliance comment:** Evidence needed: the CFO's confirmation of D30; how the K39.76 bn provision of 2025 was arrived at; the monthly Mega Farm books.
- **Where to see it:** Governance Centre: /eir-governance; Data Foundation, Loan Book (2025-11)
- **Governance setting:** mega_farms_scope; megafarm_pd_method; megafarm_scalar_ceiling
- **Reviewer sign-off:** _____ **Date:** _____